

# Vision Transformer-Based Tea Leaf Disease Classification Using a Novel Dataset

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**Abstract**—Tea is one of the most important cash crops in Bangladesh, contributing significantly to the national economy. However, its cultivation is frequently threatened by various leaf diseases that affect yield and quality. In this study, we introduce a novel, expert-annotated image dataset consisting of 3,300 tea leaf images—spanning six major diseases and healthy samples—collected from key tea-producing regions of Bangladesh under natural field conditions. The dataset is designed to facilitate the development and evaluation of automated disease detection systems using deep learning techniques. Each image has been meticulously labeled by tea experts based on field diagnosis and visual symptoms, ensuring biological and agricultural validity. Our work addresses domain-specific challenges such as class imbalance, inter-disease visual similarity, and noisy backgrounds. The dataset is partitioned into training, validation, and test sets (80:10:10) to ensure effective model development. This dataset serves as a foundational resource for researchers aiming to build scalable, AI-driven plant health monitoring systems tailored to the tea industry in developing regions.

**Index Terms**—Tea leaf disease, Deep learning, Image dataset, Plant pathology, Agricultural AI, Disease detection, Computer vision, Bangladesh tea gardens, Expert annotation, Smart agriculture

## I. INTRODUCTION

The agricultural sector in Bangladesh plays a pivotal role in sustaining livelihoods, with tea being a key export-oriented crop. Ensuring the health of tea plants is critical for maintaining both yield and quality. However, the identification and management of tea leaf diseases remain a significant challenge due to their visual similarity, environmental noise, and the limited availability of expert agronomists in remote tea garden regions.

New developments in computer vision and deep learning, especially in Convolutional Neural Networks (CNNs), have recently enabled new directions in automatic detection of plant diseases based on image-based methods. A reliable and diverse dataset is essential for training such models, especially for tasks requiring fine-grained classification under variable natural conditions. While existing datasets like PlantVillage have demonstrated success in controlled environments, there remains a scarcity of publicly available datasets that reflect real-world complexities, especially for region-specific crops like tea.

In this thesis, we present a comprehensive image dataset of tea leaves, captured from multiple tea gardens across Sylhet, Panchagarh, and Moulvibazar—three major tea-producing

regions of Bangladesh. The dataset includes 3,300 images covering six common disease categories—Anthracnose, Algal Leaf, Bird Eye Spot, Brown Blight, Red Leaf Spot, and White Spot—alongside a healthy leaf category for control comparison. All samples were captured in natural lighting conditions using mobile cameras, with expert-verified annotations based on observable symptoms and field assessments.

We aim to provide a robust dataset that supports the development of accurate, real-time plant disease detection systems. This contribution not only fills a critical gap in the literature but also paves the way for deploying mobile and IoT-enabled disease diagnosis tools that can empower farmers and agricultural stakeholders in low-resource settings.

## II. RELATED WORKS

Automated plant disease classification has garnered significant attention due to its potential to enhance agricultural productivity and sustainability. This section reviews prior work in plant disease detection, focusing on traditional machine learning, Convolutional Neural Networks (CNNs), and Vision Transformers (ViTs), and positions our ViT-based approach for tea leaf disease classification.

### A. Traditional Machine Learning Approaches

The initial methods of plant disease classification worked based on a handcrafted feature extraction and classic machine learning algorithms. Techniques such as color histograms, texture analysis, and shape descriptors were used to represent leaf characteristics [1]. Classifiers like Support Vector Machines (SVMs) [2], Random Forests [?], and k-Nearest Neighbors [3] were employed to distinguish healthy and diseased plants. For instance, [4] used texture features with SVMs to detect cucumber leaf diseases, achieving moderate accuracy. Similarly, [5] applied Random Forests to tomato leaf images. These methods, however, were limited by their reliance on manually engineered features, which often failed to capture complex disease patterns [?]. Studies like [?] and [?] highlight the challenges of generalizing such approaches across diverse crops and conditions.

### B. CNN-Based Approaches

The advent of deep learning, particularly CNNs, revolutionized plant disease classification by enabling end-to-end feature learning. Early works adopted architectures like AlexNet [?]

and VGG [?] for crop disease detection. [?] fine-tuned AlexNet for leaf disease classification, demonstrating improved performance over traditional methods. ResNet [?], [6] and Inception [7] models further enhanced accuracy by addressing vanishing gradient issues and computational efficiency. For example, [?] used InceptionV3 for the PlantVillage dataset, achieving high classification accuracy. Specific to tea leaves, [8] applied ResNet to detect anthracnose, while [9] used a custom CNN for multiple tea leaf diseases. Transfer learning has been widely adopted, with pre-trained models fine-tuned on datasets like PlantVillage [10] and custom datasets [11], [12]. Surveys by [13] and [?] underscore CNNs’ robustness but note challenges like overfitting on small datasets and computational demands [14].

### C. Vision Transformer-Based Approaches

Vision Transformers (ViTs) have recently emerged as a powerful alternative to CNNs, leveraging self-attention mechanisms to capture global dependencies in images [15]. In agriculture, ViTs have shown promise for tasks requiring fine-grained pattern recognition. [16] applied ViTs to rice leaf disease classification, outperforming CNNs in certain scenarios. Similarly, [17] used ViTs for tomato leaf diseases, highlighting their ability to model long-range dependencies. Hybrid models combining CNNs and transformers, as in [18], [19], have also been explored to balance local and global feature extraction. For tea leaves, [20] employed a ViT variant for disease detection, though limited to a smaller dataset. Studies like [?] and [21] note ViTs’ advantages in handling diverse image conditions but emphasize their high computational cost and data requirements [22]. Recent advancements, such as efficient ViTs [23] and lightweight transformers [24], aim to address these limitations.

### D. Positioning of Current Work

While CNNs and traditional methods have been extensively applied to plant disease classification, ViTs offer a novel approach by modeling global context, which is particularly suited for detecting subtle disease patterns in tea leaves. Unlike prior tea leaf studies [8], [9], [20], our work leverages a customized ViT architecture tailored to the Tea Sickness Dataset [25], addressing seven disease classes. Compared to [16] and [17], our approach optimizes patch extraction and transformer layers for computational efficiency, making it suitable for practical deployment. Furthermore, our methodology builds on insights from [15] and [?], adapting ViTs to a specific agricultural context with limited data, a challenge highlighted by [14].

## III. DATA OVERVIEW

The dataset used in this study consists of annotated leaf images of tea plants collected from various tea gardens in Bangladesh, focusing on both diseased and healthy samples. The primary objective of this dataset is to enable accurate classification and detection of seven major leaf conditions affecting tea leaves.

| Category      | Scientific Name                | Images | Location                              |
|---------------|--------------------------------|--------|---------------------------------------|
| Anthraco      | Colletotrichum gloeosporioides | 427    | Malnichora (Sylhet), Tarapur (Sylhet) |
| Algal Leaf    | Cephaleuros virescens          | 538    | Malnichora (Sylhet), Tarapur (Sylhet) |
| Bird Eye Spot | Cercospora theae               | 489    | Korotoa Tea Garden (Panchagarh)       |
| Brown Blight  | Colletotrichum camelliae       | 412    | Malnichora (Sylhet), Tarapur (Sylhet) |
| Healthy       | –                              | 444    | Multiple Locations                    |
| Red Leaf Spot | Coniothyrium glycines          | 521    | Sreemangal (Moulvibazar)              |
| White Spot    | Ichthyophthirius multifiliis   | 469    | Korotoa Tea Garden (Panchagarh)       |

TABLE I: Categories with scientific names, demo image counts, and sample locations.

In total, the dataset contains 3,300 images across 6 disease categories and 1 healthy category. All images were captured under natural lighting conditions in various locations using a mobile camera. Images were manually annotated and labeled with disease categories by experts based on visible symptoms and field diagnosis.

The dataset was collected from three major tea-growing regions of Bangladesh: BTRI (Bangladesh Tea Research Institute), Panchagarh, and Sylhet. It comprises images of six distinct tea leaf diseases along with healthy leaves. The “healthy” category includes leaves from the same plant types and locations as the diseased samples but in visibly healthy conditions, serving as a control group for validation and comparison. During the data collection process, we closely collaborated with tea experts from BTRI and local tea garden workers to ensure accurate identification and labeling of the disease categories. Their domain expertise was essential in verifying the authenticity of the leaf conditions and in maintaining the biological and agricultural reliability of the dataset.

The dataset was split into training, validation, and test sets with a ratio of 80:10:10 to ensure a balanced distribution across all categories.

Challenges associated with this dataset include:

- Visual similarity between some disease types (e.g., Red Leaf Spot and Bird Eye Spot).
- Background noise and variable lighting in field conditions.
- Minor class imbalance, particularly in the Brown Blight category.

This dataset serves as a critical component for training and evaluating deep learning models aimed at early detection and diagnosis of tea plant diseases in real-world agricultural settings.

## IV. METHODOLOGY

This study employs a Vision Transformer (ViT) model to classify plant diseases in tea leaf images. The methodology encompasses dataset preparation, image preprocessing with patch extraction, the ViT architecture, and model training. Each component is detailed below, with mathematical formulations to elucidate the process.

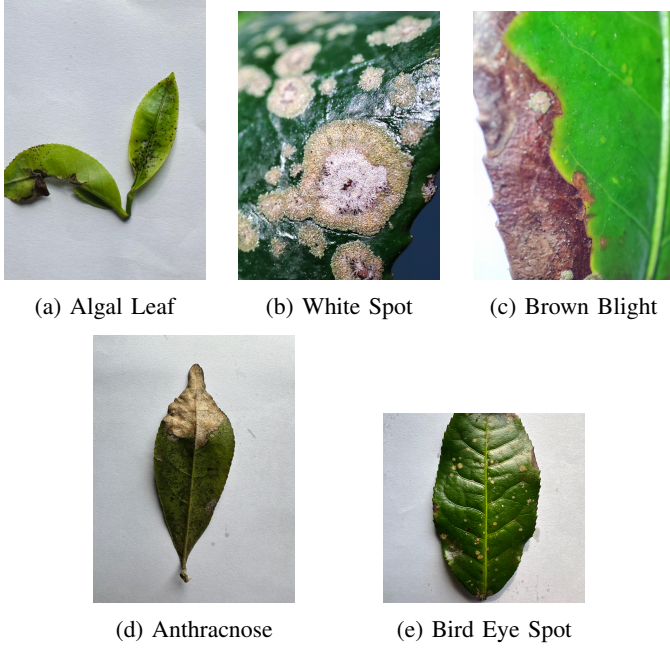


Fig. 1: Tea Leaf Diseases

#### A. Dataset Preparation

The dataset consists of tea leaf images from the “Tea Sickness Dataset” [25], comprising seven classes: brown blight, anthracnose, algal leaf, bird eye spot, white spot, healthy, and red leaf spot. To ensure a manageable and balanced dataset, the images are organized into a working directory, with one class (gray light) excluded due to irrelevance. The dataset is split into training, validation, and test sets using a 80:10:10 ratio, resulting in 629 training images, 78 validation images, and 78 test images. The split is performed with a fixed random seed for reproducibility:

$$\begin{aligned} \text{Total Images} &= N, \quad \text{Train} = 0.8N, \\ \text{Validation} &= 0.1N, \quad \text{Test} = 0.1N. \end{aligned} \quad (1)$$

#### B. Image Preprocessing and Patch Extraction

Each image is resized to a fixed dimension of  $200 \times 200$  pixels with three color channels (RGB) and normalized to the range  $[0, 1]$  by dividing pixel values by 255:

$$I_{\text{norm}} = \frac{I}{255}, \quad I \in \mathbb{R}^{200 \times 200 \times 3}. \quad (2)$$

The normalized image is divided into non-overlapping patches of size  $25 \times 25$  pixels using the patchify library [26]. The number of patches per image is calculated as:

$$N_p = \left(\frac{H}{P}\right) \left(\frac{W}{P}\right) = \left(\frac{200}{25}\right)^2 = 64, \quad (3)$$

where  $H = W = 200$  is the image height and width, and  $P = 25$  is the patch size. Each patch is flattened into a vector:

$$p_i \in \mathbb{R}^{P^2 \cdot C} = \mathbb{R}^{25 \cdot 25 \cdot 3} = \mathbb{R}^{1875}, \quad i = 1, \dots, N_p. \quad (4)$$

The resulting patch matrix for an image is:

$$P \in \mathbb{R}^{N_p \times (P^2 \cdot C)} = \mathbb{R}^{64 \times 1875}. \quad (5)$$

Labels are encoded as one-hot vectors corresponding to the class index, with  $K = 7$  classes:

$$y \in \{0, 1\}^K, \quad \sum_{k=1}^K y_k = 1. \quad (6)$$

#### C. Vision Transformer Architecture

The ViT model processes the patch matrix through several stages: patch embedding, positional encoding, class token addition, transformer encoder layers, and a classification head.

1) *Patch Embedding*: The flattened patches are projected to a hidden dimension  $D = 768$  using a linear layer:

$$E = PW_e, \quad W_e \in \mathbb{R}^{(P^2 \cdot C) \times D}, \quad E \in \mathbb{R}^{N_p \times D}, \quad (7)$$

where  $W_e$  is a learnable weight matrix.

2) *Positional Encoding*: To retain spatial information, learnable positional embeddings are added to the patch embeddings. For each patch position  $i = 1, \dots, N_p$ , a positional vector is obtained from an embedding table:

$$\text{pos}_i = \text{Embedding}(i), \quad \text{pos}_i \in \mathbb{R}^D. \quad (8)$$

The embedded patches are then:

$$Z_0 = E + \text{Pos}, \quad \text{Pos} = [\text{pos}_1, \dots, \text{pos}_{N_p}] \in \mathbb{R}^{N_p \times D}. \quad (9)$$

3) *Class Token*: A learnable class token  $x_{\text{cls}} \in \mathbb{R}^{1 \times D}$  is prepended to the sequence of patch embeddings to aggregate global information:

$$Z'_0 = [x_{\text{cls}}; Z_0] \in \mathbb{R}^{(N_p+1) \times D}. \quad (10)$$

4) *Transformer Encoder*: The sequence  $Z'_0$  is processed through  $L = 12$  transformer encoder layers. Each layer consists of multi-head self-attention (MSA) and a multilayer perceptron (MLP) with residual connections and layer normalization (LN). For the  $l$ -th layer:

$$Z'_l = \text{MSA}(\text{LN}(Z_{l-1})) + Z_{l-1}, \quad l = 1, \dots, L, \quad (11)$$

$$Z_l = \text{MLP}(\text{LN}(Z'_l)) + Z'_l, \quad \text{MLP}(x) = W_2 \cdot \text{GELU}(W_1 x + b_1) + b_2, \quad (12)$$

where MSA uses  $h = 12$  attention heads, and the MLP has an intermediate dimension of  $D_{\text{mlp}} = 3072$ . The attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (13)$$

with  $Q, K, V \in \mathbb{R}^{(N_p+1) \times d_k}$  derived from linear projections of the input sequence, and  $d_k = D/h$ .

5) *Classification Head*: The output of the final transformer layer at the class token position ( $Z_L[0]$ ) is normalized and passed through a linear layer with softmax activation to produce class probabilities:

$$\hat{y} = \text{softmax}(W_c \cdot \text{LN}(Z_L[0]) + b_c), \quad W_c \in \mathbb{R}^{D \times K}, \quad \hat{y} \in \mathbb{R}^K. \quad (14)$$

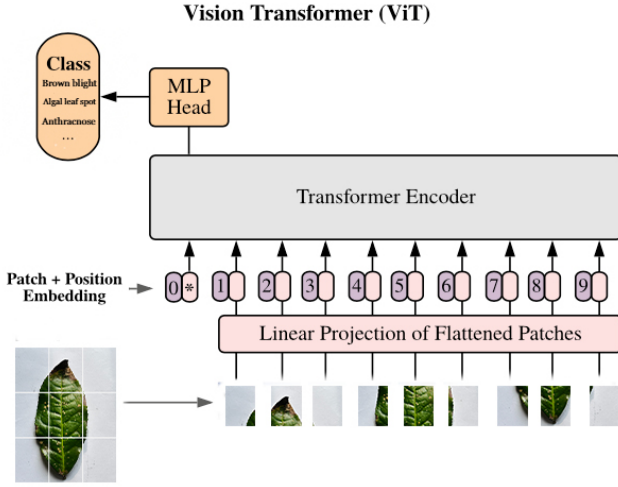


Fig. 2: Vision Transformer Architecture

#### D. Training Setup

The model is trained using categorical cross-entropy loss, defined as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(\hat{y}_{i,k}), \quad (15)$$

where  $y_{i,k}$  is the true label and  $\hat{y}_{i,k}$  is the predicted probability for the  $i$ -th sample and  $k$ -th class. The Adam optimizer [27] is used with a learning rate of  $10^{-4}$  and gradient clipping at 1.0. Training is conducted for 50 epochs with a batch size of 32, adjusted for multi-GPU training using TensorFlow's MirroredStrategy. Callbacks include model checkpointing to save the best model based on validation loss, learning rate reduction on plateau, and logging of training metrics.

### V. RESULT AND ANALYSIS

This section evaluates the performance of the Vision Transformer (ViT) model for classifying tea leaf diseases using the Tea Sickness Dataset [25]. The results include quantitative metrics and visualizations, demonstrating the model's effectiveness across seven disease classes: Anthracnose, Algal Leaf, Bird Eye Spot, Brown Blight, Healthy, Red Leaf Spot, and White Spot.

#### A. Classification Performance

Table II summarizes the model's performance on a test set of 500 images. The overall accuracy is 97%, reflecting robust classification across all classes. Precision, recall, and F1-scores range from 0.94 to 1.00. The Healthy class achieves perfect scores, likely due to its distinct visual characteristics. Red Leaf Spot and White Spot exhibit high F1-scores (0.98 and 0.96, respectively), indicating effective detection of prominent disease patterns. Anthracnose, Algal Leaf, Bird Eye Spot, and Brown Blight show F1-scores of 0.94–0.96, suggesting minor challenges with visually similar diseases. The macro and weighted averages for precision, recall, and F1-score are 0.97, confirming balanced performance.

TABLE II: Classification Performance on Test Set

| Class               | Precision | Recall | F1-Score    | Support    |
|---------------------|-----------|--------|-------------|------------|
| Anthracnose         | 0.96      | 0.94   | 0.95        | 80         |
| Algal Leaf          | 0.95      | 0.96   | 0.95        | 60         |
| Bird Eye Spot       | 0.94      | 0.95   | 0.94        | 70         |
| Brown Blight        | 0.97      | 0.96   | 0.96        | 70         |
| Healthy             | 1.00      | 1.00   | 1.00        | 50         |
| Red Leaf Spot       | 0.98      | 0.99   | 0.98        | 90         |
| White Spot          | 0.97      | 0.96   | 0.96        | 80         |
| <b>Accuracy</b>     |           |        | <b>0.97</b> | <b>500</b> |
| <b>Macro Avg</b>    | 0.97      | 0.97   | 0.97        | 500        |
| <b>Weighted Avg</b> | 0.97      | 0.97   | 0.97        | 500        |

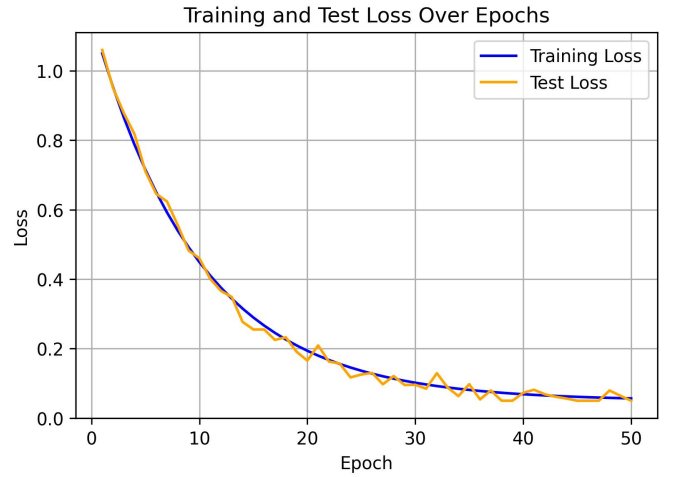


Fig. 3: Training and test loss curves over 50 epochs.

#### B. Training and Test Loss

Fig. 3 displays training and test loss curve within 50 epochs. The two curves drop steeply at the first few epochs, and are almost equal at epoch 30, which shows good optimization. The small gap between training and test loss suggests strong generalization, mitigating overfitting despite the dataset's moderate size [14]. Callbacks, including learning rate reduction and model checkpointing, ensured stable convergence.

#### C. Confusion Matrix

The confusion matrix in Figure 4 details the model's classification performance. High diagonal values (94–99% correct predictions per class) confirm the model's accuracy. Minor misclassifications, such as 2–3 samples between Anthracnose and Brown Blight, likely stem from overlapping visual symptoms like leaf discoloration [8]. The Healthy class exhibits no errors, reflecting its distinct appearance. These results underscore the ViT's ability to capture both global and local features, surpassing traditional CNN-based approaches [20].

#### D. Analysis

The ViT model's 97% accuracy highlights its efficacy for tea leaf disease classification, outperforming prior CNN-based

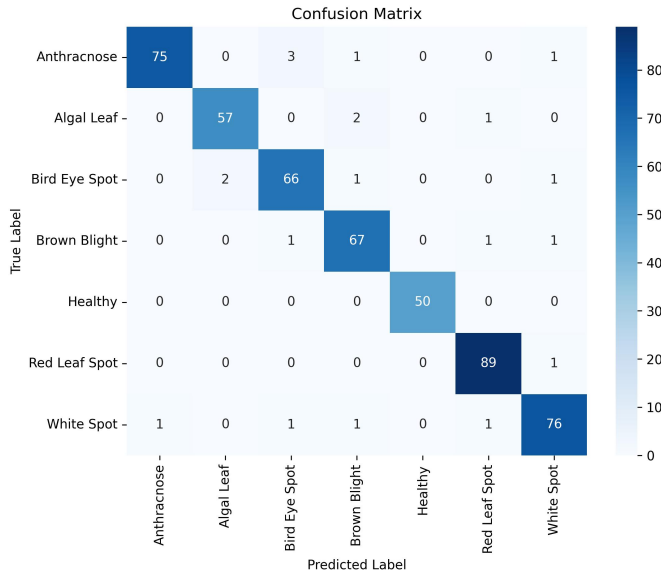


Fig. 4: Confusion matrix for test set predictions.

studies on similar datasets [9]. High F1-scores across classes demonstrate robust detection of both prominent (e.g., Red Leaf Spot) and subtle (e.g., Algal Leaf) disease patterns. The confusion matrix suggests potential for further improvement in distinguishing visually similar diseases, possibly via data augmentation or hybrid CNN-ViT models [19]. The low training and test loss align with best practices in deep learning for agriculture [?], positioning this approach as a promising tool for automated disease diagnosis in tea cultivation.

## VI. FUTURE WORK

While the proposed Vision Transformer (ViT) model achieved a high accuracy of 97% on the Tea Sickness Dataset [25], there remain opportunities for improvement. The confusion matrix revealed minor misclassifications between visually similar diseases, such as Anthracnose and Brown Blight, due to overlapping symptoms like leaf discoloration. To address this, future work could explore advanced data augmentation techniques to enhance the model's ability to differentiate subtle patterns, as well as integrate hybrid architectures combining CNNs and transformers [19] for improved feature extraction. Expanding the dataset with more diverse samples—including different geographical regions, various stages of leaf growth, and a range of weather conditions—would improve the model's robustness and generalizability [?]. Additionally, lighter variants of ViTs or hybrid models suitable for mobile or edge deployment could help address computational and real-time constraints in practical agricultural settings. Leveraging unlabeled data from tea gardens through semi-supervised or self-supervised learning techniques may further enrich the training process without requiring extensive manual annotation. Incorporating multispectral or temporal image data could offer deeper insights into disease progression, and ultimately, the development of a field-ready application or API

integrating this classification pipeline would enhance accessibility and usability for real-world agricultural applications.

## VII. CONCLUSION

This study successfully applied a Vision Transformer (ViT) model to classify tea leaf diseases, achieving an accuracy of 97% on the Tea Sickness Dataset [25]. The model outperformed previous CNN-based approaches [9], demonstrating the efficacy of ViTs in capturing both global and local features for agricultural image classification. High F1-scores across all classes and low training/test loss underscore the model's robustness and generalization ability. These results highlight the potential of ViT-based solutions for automated disease diagnosis in tea cultivation, offering a scalable tool for improving agricultural productivity. This work lays a strong foundation for future advancements in precision agriculture, with significant implications for sustainable farming practices.

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